Engineering Mathematics - I · Unit_01 · Unit 1

DRAFT — UE26MA141A · Unit_01 · Unit 1 · Extremas of functions

Standalone video lesson · authored study notes with source citations

Hugging Face PESU_Materials source tree, official units.json catalogue, transcript sidecars, slide text/PDF/PPT sources and the selected First_Year_8enterprise archive; see the provenance datasets in the release ZIP. Generated 18 October 2026.

· How to use this pack

Read the topic explanation, worked examples, and recap. Evidence citations point to source transcript intervals and slide pages; full slide images remain in the source visual index. Topics include the applicable parts: overview, key terms, formulas, figures, flowcharts, important points, worked examples, past-paper questions and a recap. IDs are shared with the separate documents: E = formula, G = graph, F = flowchart, W = worked example. Each ID is numbered by topic, so E4.2 is the second formula of topic 4.

Source text	Verbatim or OCR-derived text, kept separate from editorial notes.
Candidate	Automated topic/source association requiring manual confirmation.
Low ASR confidence	Transcript segment requires audio verification.
DRAFT	Human review remains incomplete.

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M1

Extremas of functions

1 Extremas of functions

Study focus: Extremas of functions. This topic explains the central idea, its conditions, and how the lecture applies it in context. Review the selected source statements and follow their citations for the full lesson segment. EV0000194 · EV0000195

SHORT-NOTE RECAP

Core explanation · 00:00–04:59

Today we'll be doing applications of partial differentiation, which is extremes of functions.

Key details

- I hope you remember how you calculated maxima for a function of single variable, right?
- And then we wanted to know what are these points, right?
- That's what we did.
- It could be either maximum or minimum.
- The first thing I did was to calculate the critical points.
- You calculate the first derivative and equate them to zero.
- What is the practical advantage of this?
- that if you have a local maxima, then the tangent at these points will be parallel to the x axis.
- Can you visualize that here?
- See k you can see that here.
- So we get the critical point by solving this equation that's the first step.
- The second step would be you've got to find the second order partial derivative.
- Second order not partial derivative, second order derivative.
- And if the value of this is less than zero,.
- At the critical point, we say that the function is maximum, right?
- And if it is greater than zero we say that the function has a minimum at that point.
- And if t square y by dx square is equal to zero, then it's neither maximum nor minimum.
- And the condition holds for both because the tangents as you see, the tangents at these points are also parallel to the x axis.
- So basically what we did you calculate the first order derivative, equate it to zero to get the.

- Critical points means the points at which the curve might have a extrema, either it could be a maxima or a minima.
- Then we need to decide whether that critical point represents a maximum point or a minimum point.
- If it gives zero then we cannot conclude anything.
- Such points are called saddle point.
- I hope you remember that, right so a saddle point looks something like this where so if saddle point is this where the function is neither maximum nor minimum.
- That means the test fails basically.
- But before that when do we say that a function of two independent variables has a maximum?
- I can decide that this point is the maximum for this surface at a point A, comma b if.
- Any neighborhood point the value of the function is less than the value of the function at A B.
- That means this condition.
- If f of ab is greater than f of a plus h, comma b plus k.
- Neighborhood means very close to this.
- Then we say that the function has a relative maximum.
- Why the word relative?
- It's because I'm considering the points that are closer to A B, I'm in a finite range, right?
- I'm not taking the surface in the entire real plane from minus infinity to infinity.
- That's why the term relative maximum.
- Because in that neighborhood that's the maximum point.

Example or application from the lecture

- Suppose that's the case.
- Or suppose I have a curve or a surface something like this.
- And let's say I have a point here and I call that as f of x comma y.

These are edited study notes based on the cited lecture interval. Check uncertain speech recognition against the source transcript and recording.

EV0000194 · EV0000195 · EV0000196 · +1 source units in pack.json

Core explanation · 05:00–09:59

Then we say that's a relative minimum.

Key details

- So that means.
- In other words, you can have another definition if delta of A plus H comma B plus K minus F of A B is of the same sign, that's what I mean.
- So you just look at this condition here.
- For all small h and k and if sin is negative then f of ab is maximum.
- If it's positive we say that f of ab is minimum.
- That's the definition, okay.

- So you don't need to worry too much about it because what we call it as a maximum point and what we define it as a minimum point.
- So what we did first is we got dy by dx equal to 0.
- So, corresponding to this, what we have for a function of two independent variables, that there are two first order partial derivatives.
- So the first one to find the critical point I'm going to equate the first order partial derivatives to 0.
- And the second condition, what is the second condition here?
- I need to find.
- The second order derivative, substitute the point that you got here and check if the value is negative or if it's positive or if it's equal to C.
- That is, I am going to find because that there are four mixed order partial derivatives, right?
- R T minus a s square.
- Where what is r?
- R is nothing but f x.
- And so F y is also y x is also S.
- that mixed second order partial derivatives are always equal and FYY will be T.
- And this determinant is called a Hesian matrix.
- Is that's a condition I want.
- So remember here we got it directly, but here we need to check another condition.
- So the first preliminary condition that should be satisfied for a point to represent an extrema, it could be a maxima or a minima is f x into f y minus f x y the whole square must be greater than zero.
- If this R T minus S square is less than 0, then f of a b is not an extrema value at all.
- You cannot conclude anything from that.
- Just in case if it becomes equal to zero, then the test completely fails.
- You don't know whether you don't know it the you don't get any information about the nature of that point, whether it is a maxima or a minima or a saddle point, nothing we get.
- The first thing is find out both the first order partial derivatives, equate them to zero.
- Secondly, find R T minus S square, that is a combination of the second order partial derivative.
- At some point if this is zero, how do I decide that point is a maximum or a minimum.
- In that case, we have few conditions.
- Along with this condition, if R is less than 0 at any point AB, then f of A B has a maximum value.
- R means what then?
- This must be greater than 0.
- Along with this you need to check that f x must be less than 0.

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EV0000198 · EV0000199 · EV0000200 · +1 source units in pack.json

Core explanation · 10:00–14:59

which is R.

Key details

- So if this condition is satisfied then it will have a minimum or maximum.
- This is similar to the condition, see you remember this d square y by d x square has to be less than 0.
- And the second condition is so if R T minus S square this has to hold good for it to be an extrema.
- Along with that I have F Y now to be.
- sorry F x only.
- So, for a condition for it to be minimum now, right?
- Condition for it to be minimum is f x must be greater than 0 and that's what you concluded, right?
- In the case of single variable.
- If d square y by dx square is greater than 0, then the function will have a minimum at that point.
- And then lastly if I have R T minus S square to be less than 0, then it's it represents a saddle point.
- It's it doesn't sway to the left, neither it sway to the right.
- And of course if this is zero then I cannot conclude anything, so you have to say that no conclusion can be made.
- The first thing is you find out f x equated to zero, find f y equated to zero.
- Solve these two equations to get critical points.
- Next is you calculate the second order partial derivative.
- This has to be positive.
- Along with this, if f x is less than 0, I'll get maximum.
- That means that point represents maximum.
- And if x f x is greater than 0,.
- And if f x into f y minus fxy the whole square, if it is less than 0, then it's a saddle point.
- And if just in case if this is equal to zero then no conclusion.
- Is it clear?

These are edited study notes based on the cited lecture interval. Check uncertain speech recognition against the source transcript and recording.

EV0000202 · EV0000203 · EV0000204 · +1 source units in pack.json

A Symbol glossary

Symbol	Name	Meaning	Appears in

B Sources, coverage and gaps

Source	Path	Use
Extremas of functions transcript (JSON)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_01/AV_Summaries/009_Extremas_of_functions__5ebc2dd46f817380.transcript.json	Full JSON sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.
Extremas of functions transcript (Markdown)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_01/AV_Summaries/009_Extremas_of_functions__5ebc2dd46f817380.transcript.md	Full Markdown sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.
Extremas of functions transcript (plain text)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_01/AV_Summaries/009_Extremas_of_functions__5ebc2dd46f817380.transcript.txt	Full plain text sidecar. Normalized text similarity to canonical JSON: 0.9993; only the JSON segmentation supplies the displayed timestamps.
Extremas of functions transcript (SRT)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_01/AV_Summaries/009_Extremas_of_functions__5ebc2dd46f817380.srt	Full SRT sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.

Historical question-source excerpts are shown only when a page has a strong text-based unit match. They are source references, not new solutions or exam predictions.

Known gaps, conflicts and corrections

1. No historical archive question page met the unit-match candidate threshold; all subject-level and ambiguous pages remain in datasets/question-archive.json.
2. 2 video-to-slide associations were not promoted to a match. Candidate rankings are listed in the transcript corpus and video-slide-matches.csv; the slide decks appear separately.
3. 1 transcript segments have low ASR confidence; they are flagged with timestamps in the transcript blocks and indexed in the corpus.
4. This is a visibly labelled source-preservation draft. Human page-level semantic and visual review is not complete; no unsupported subject summary or formula correction is claimed.

5. The 19.26 GiB MP4 files remain hosted on Hugging Face; the full video path/hash and transcript coverage are indexed. The full audio/video was not watched end-to-end.
6. Formula candidates are machine-extracted text with source locators; use the adjacent original slide page and the formula index before treating a candidate as an authoritative equation.
7. The study PDF contains edited, topic-organized lecture notes and textual slide-page takeaways. Full-page slide images remain in the source/visual index datasets, not embedded as the lesson. Human content, ASR, formula, and page-level visual review remains pending.

Coverage

Evidence is summarized by source to keep the reading copy legible. Full source-unit IDs, locators and decisions remain in pack.json.

Source	Unit counts	Example decisions
SRC00050	3 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0021. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0022 to s0040. Sidecar variants compare... t000600-000899: Timestamp interval t000600-000899; canonical segment IDs s0041 to s0052. Sidecar variants compare...

Source	Unit counts	Example decisions
SRC00051	3 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0021. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0022 to s0040. Sidecar variants compare... t000600-000899: Timestamp interval t000600-000899; canonical segment IDs s0041 to s0052. Sidecar variants compare...
SRC00052	3 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0021. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0022 to s0040. Sidecar variants compare... t000600-000899: Timestamp interval t000600-000899; canonical segment IDs s0041 to s0052. Sidecar variants compare...

Source	Unit counts	Example decisions
SRC00053	3 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0021. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0022 to s0040. Sidecar variants compare... t000600-000899: Timestamp interval t000600-000899; canonical segment IDs s0041 to s0052. Sidecar variants compare...

C Revision and retrieval practice